

When Does Aggregating Explanations Work?

TissueMNIST Results

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This report is generated from the structured result files listed in `RESULT_MANIFEST.json`. It is organised by model and experiment and retains incomplete and historical snapshots as archival tables.

Historical and superseded result tables

A DenseNet-121 (archived)

Table 1: Real-checkpoint explainer compatibility pilot for TissueMNIST–DenseNet-121. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	finite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	DenseNet-121	passed	blocked	Saliency	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	DenseNet-121	passed	blocked	InputXGradient	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	DenseNet-121	passed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0919626.6774e+09		
TissueMNIST	DenseNet-121	passed	blocked	GuidedBackprop	true	true	1	true	1				
TissueMNIST	DenseNet-121	passed	blocked	Deconvolution	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	DenseNet-121	passed	blocked	FeatureAblation	true	true	1	true	1				
TissueMNIST	DenseNet-121	passed	blocked	Occlusion	true	true	1	true	1				
TissueMNIST	DenseNet-121	passed	blocked	DeepLift	true	true	1	true	1	1	0.0582596.5947e+08		
TissueMNIST	DenseNet-121	passed	blocked	GradientShap	true	true	1	true	1				
TissueMNIST	DenseNet-121	passed	blocked	DeepLiftShap	true	true	1	true	1	1	0.0528091.7365e+09		
TissueMNIST	DenseNet-121	passed	blocked	LRP	true	true	1	true	1				

B EfficientNet-B0 (archived)

Table 2: Real-checkpoint explainer compatibility pilot for TissueMNIST-EfficientNet-B0. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	infinite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	EfficientNet-B0	Failed	blocked	Saliency	true	true	1	true	1	1	0.2524253	8024e+08	
TissueMNIST	EfficientNet-B0	Failed	blocked	InputXGradient	true	true	1	true	1	1	0.1812333	8265e+08	
TissueMNIST	EfficientNet-B0	Failed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0371914	5561e+09	
TissueMNIST	EfficientNet-B0	Failed	blocked	GuidedBackprop	true	true	1	true	1				
TissueMNIST	EfficientNet-B0	Failed	blocked	Deconvolution	true	true	1	true	1				
TissueMNIST	EfficientNet-B0	Failed	blocked	FeatureAblation	true	true	1	true	1				
TissueMNIST	EfficientNet-B0	Failed	blocked	Occlusion	true	true	1	true	1				
TissueMNIST	EfficientNet-B0	Failed	blocked	DeepLift	true	true	1	true	1	1	0.0232442	7867e+08	
TissueMNIST	EfficientNet-B0	Failed	blocked	GradientShap	true	true	1	true	1				
TissueMNIST	EfficientNet-B0	Failed	blocked	DeepLiftShap	true	true	1	true	1	1	0.0226736	5413e+08	
TissueMNIST	EfficientNet-B0	Failed	blocked	LRP	false								RuntimeError: No expla- nation batch candi- date passed

C MobileNetV3-Large (archived)

Table 3: Real-checkpoint explainer compatibility pilot for TissueMNIST–MobileNetV3-Large. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	infinite fraction	target sensitive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	MobileNetV3-Large	failed	blocked	Saliency	true	true	1	true	1				
TissueMNIST	MobileNetV3-Large	failed	blocked	InputXGradient	true	true	1	true	1				
TissueMNIST	MobileNetV3-Large	failed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0245642	4.103e+09	
TissueMNIST	MobileNetV3-Large	failed	blocked	GuidedBackprop	true	true	1	true	1				
TissueMNIST	MobileNetV3-Large	failed	blocked	Deconvolution	true	true	1	true	1				
TissueMNIST	MobileNetV3-Large	failed	blocked	FeatureAblation	true	true	1	true	1				
TissueMNIST	MobileNetV3-Large	failed	blocked	Occlusion	true	true	1	true	1				
TissueMNIST	MobileNetV3-Large	failed	blocked	DeepLift	true	true	1	true	1	1	0.0209562	2.835e+08	
TissueMNIST	MobileNetV3-Large	failed	blocked	GradientShap	true	true	1	true	1				
TissueMNIST	MobileNetV3-Large	failed	blocked	DeepLiftShap	true	true	1	true	1	1	0.0214065	5.0209e+08	
TissueMNIST	MobileNetV3-Large	failed	blocked	LRP	false								RuntimeError: No explanation batch candidate passed

D ResNet-18 (archived)

Table 4: Real-checkpoint explainer compatibility pilot for TissueMNIST-ResNet-18. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	finite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	ResNet-18	passed	blocked	Saliency	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	passed	blocked	InputXGradient	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	passed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0352721	2721.2722e+09	
TissueMNIST	ResNet-18	passed	blocked	GuidedBackprop	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	passed	blocked	Deconvolution	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	passed	blocked	FeatureAblation	true	true	1	true	1				
TissueMNIST	ResNet-18	passed	blocked	Occlusion	true	true	1	true	1				
TissueMNIST	ResNet-18	passed	blocked	DeepLift	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-18	passed	blocked	GradientShap	true	true	1	true	1				
TissueMNIST	ResNet-18	passed	blocked	DeepLiftShap	true	true	1	true	1	1	0.0169024	24.4585e+08	
TissueMNIST	ResNet-18	passed	blocked	LRP	true	true	1	true	1				

E ResNet-50 (archived)

Table 5: Real-checkpoint explainer compatibility pilot for TissueMNIST-ResNet-50. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	finite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	ResNet-50	passed	blocked	Saliency	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	passed	blocked	InputXGradient	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	passed	blocked	IntegratedGradients	true	true	1	true	1	1	0.0739514	5.233e+09	
TissueMNIST	ResNet-50	passed	blocked	GuidedBackprop	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	passed	blocked	Deconvolution	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	passed	blocked	FeatureAblation	true	true	1	true	1				
TissueMNIST	ResNet-50	passed	blocked	Occlusion	true	true	1	true	1				
TissueMNIST	ResNet-50	passed	blocked	DeepLift	false	false	1		1				RuntimeError: Deterministic repeat check failed
TissueMNIST	ResNet-50	passed	blocked	GradientShap	true	true	1	true	1				
TissueMNIST	ResNet-50	passed	blocked	DeepLiftShap	true	true	1	true	1	1	0.0261371	1.2474e+09	
TissueMNIST	ResNet-50	passed	blocked	LRP	true	true	1	true	1				

F ViT-B/16 (archived)

Table 6: Real-checkpoint explainer compatibility pilot for TissueMNIST–ViT-B/16. This archival diagnostic reports per-method execution checks and is not a Phase 2 quality or robustness result.

dataset	model	result status	gate status	method	passed	deterministic repeat	infinite frac- tion	target sensi- tive	selected batch size	attack batch size	attack seconds per sample	attack peak cuda bytes	error
TissueMNIST	ViT-B/16	passed	passed	Saliency	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	InputXGradient	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	IntegratedGradients	true	true	1	true	1	1	0.1408036	6673e+09	
TissueMNIST	ViT-B/16	passed	passed	FeatureAblation	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	Occlusion	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	GradientShap	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	CheferTransformerAttribution	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	PartialLRP	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	FullLRP	true	true	1	true	1				
TissueMNIST	ViT-B/16	passed	passed	GradientAttentionRollout	true	true	1	true	1	1	0.0622943	3842e+09	
TissueMNIST	ViT-B/16	passed	passed	AttentionGradCAM	true	true	1	true	1	1	0.0516873	3105e+09	